

Emma Dilic

Cartographer & Geospatial Analyst

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EDUCATION

Virginia Tech, College of Natural Resources and Environment

Expected May 2027

Blacksburg, VA

Bachelor of Arts in Geography · Minor in Geographic Information Systems (GIS)

EXPERIENCE

National Geographic Cartography Intern

May 2026 – August 2026

National Geographic · Washington, D.C.

- Created print and digital maps for National Geographic editorial stories, producing three digital maps, four print maps, and 20+ sketch maps for a forthcoming Bookazine atlas.
- Designed and refined maps using ArcGIS Pro, QGIS, Adobe Illustrator, Photoshop, and MAPublisher while adhering to National Geographic cartographic standards.
- Researched and visualized geographic data for environmental, wildlife, exploration, and global storytelling projects.
- Collaborated with cartographers, text editors, graphic designers, and researchers to develop accurate and visually compelling maps for publication.

Teaching Assistant for Cartography

January 2025 – Present

Virginia Tech

- Engaged with four lab sections per week, totaling 55 students, providing direction and feedback for thematic map assignments.
- Provided constructive feedback regarding aesthetics and functionality of map projects.
- Regularly updated lab tutorial materials, ensuring up-to-date ArcGIS Pro screenshots and examples.

Teaching Assistant for Maps and Mapping

August 2024 – December 2024

Virginia Tech

- Supported course instruction by attending all lectures and labs, assisting 110 students.
- Answered questions to strengthen understanding of map reading and measurement skills.

GIS PROJECTS

Historical Map of Botetourt

February 2025 – April 2025

Botetourt Historical Society and Museum

- Partnered with the Historical Society to design a historical map of Botetourt County (circa 1776); final map displayed at the Historical Society Museum.

Determinants of Pediatric Vaccine Uptake

June 2022 – August 2022

George Mason University

- Conducted spatial analyses of CDC and county-level data on pediatric COVID-19 vaccine disparities (ASSIP); designed thematic maps in ArcGIS Pro and QGIS.

FCPS Safe Routes to School

September 2022 – June 2023

Fairfax County Public Schools · Fairfax, VA

- Designed 40 safe walking routes for elementary, middle, and high schools using GIS network and safety analysis.

RESEARCH

Advanced Cartography and Geovisualization

2025 – Present

Virginia Tech

Undergraduate research supporting the development of a new advanced cartography and geovisualization course.

Biogeographic Cartography – Association of American Geographers

Spring 2026

Virginia Tech

Designed two publication-quality maps of plant community distribution in Glacier National Park.

Medical Geography – Independent Study

Spring 2026

Virginia Tech

Independent study on malaria transmission and spatial patterns in Bangladesh.

ADDITIONAL EXPERIENCE & INVOLVEMENT

President, Geography Club

Virginia Tech · Aug 2025 – Present

Bookseller

Barnes and Noble · Mar 2023 – Dec 2025

Ambassador, College of Natural Resources

Virginia Tech · Jan 2024 – Present

Etsy Shop Owner

Self-employed · Oct 2020 – Dec 2023

TECHNICAL SKILLS & LANGUAGES

ArcGIS Pro	Advanced	English	Fluent
ArcGIS Online Suite	Advanced	Bosnian	Fluent
MAPublisher	Intermediate	Serbian	Conversational
Adobe Illustrator	Intermediate	Croatian	Conversational
Adobe Photoshop	Intermediate		
Python	Novice/Intermediate		
R	Novice/Intermediate		

AWARDS & HONORS

First Place, VGA StoryMap Competition (2023) · Submission to the CAGIS Annual Cartography Competition (2024)

PUBLICATIONS & PRESENTATIONS

Contributed two maps to an Association of American Geographers publication on plant community distribution in Glacier National Park (Biogeographic Cartography research, 2026).